

APEXPLANET SOFTWARE PVT LTD

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CYBER SECURITY INTERNSHIP

TASK 1 REPORT

FOUNDATIONS OF CYBER SECURITY AND LAB ENVIRONMENT SETUP

ABSTRACT

The objective of this task is to understand the fundamentals of cybersecurity and create a practical cybersecurity laboratory environment. The project covers cybersecurity concepts, Linux fundamentals, networking basics, cryptography basics, and the setup of a penetration testing lab using Kali Linux and Metasploitable2 in Oracle VirtualBox.

The laboratory was used to perform host discovery and service enumeration using Nmap. Additional cybersecurity tools such as Wireshark and Burp Suite were explored to understand their role in network analysis and web application security testing.

This task provides a strong foundation for advanced penetration testing and ethical hacking activities.

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1. INTRODUCTION TO CYBER SECURITY

Cyber Security refers to the practice of protecting computer systems, networks, applications, and data from cyber attacks, unauthorized access, theft, and damage.

With the increasing use of technology and the internet, organizations face numerous security threats daily. Cybersecurity helps protect sensitive information and maintain system availability.

Importance of Cyber Security

- Protects confidential information
 - Prevents cyber attacks
 - Ensures data integrity
 - Maintains business continuity
 - Secures online transactions
-

2. CIA TRIAD

The CIA Triad is the foundation of information security.

2.1 Confidentiality

Confidentiality ensures that information is accessible only to authorized individuals.

Examples:

- Passwords
- Encryption
- Access Control

2.2 Integrity

Integrity ensures that data remains accurate and unaltered.

Examples:

- Hashing
- Checksums
- Digital Signatures

2.3 Availability

Availability ensures that systems and data remain accessible whenever required.

Examples:

- Backup Systems
 - Redundancy
 - Disaster Recovery
-

3. CYBER THREATS AND ATTACK VECTORS

3.1 Malware

Malware refers to malicious software designed to damage systems.

Examples:

- Virus
- Worm
- Trojan Horse

3.2 Phishing

Phishing is a social engineering attack where attackers trick users into revealing sensitive information.

3.3 Ransomware

Ransomware encrypts user data and demands payment for decryption.

3.4 Distributed Denial of Service (DDoS)

A DDoS attack overwhelms a system with excessive traffic.

3.5 SQL Injection

SQL Injection exploits vulnerabilities in database queries to gain unauthorized access.

3.6 Brute Force Attack

Attackers repeatedly attempt passwords until the correct one is found.

Attack Vectors

Social Engineering

Manipulating users into revealing confidential information.

Wireless Attacks

Attacks targeting Wi-Fi networks and wireless devices.

Insider Threats

Security risks originating from employees or authorized users.

4. LINUX FUNDAMENTALS

Linux is a widely used operating system in cybersecurity.

Basic Commands Practiced

Present Working Directory

```
pwd
```

List Files

```
ls
```

Change Directory

```
cd
```

Create Directory

```
mkdir ApexPlanet
```

Create File

```
touch notes.txt
```

View File Permissions

```
ls -l
```

Change Permissions

```
chmod 777 notes.txt
```

User Information

```
whoami
```

System Information

```
uname -a
```

Network Information

```
ip addr
```

Connectivity Test

```
ping google.com
```

5. NETWORKING BASICS

Networking is an essential component of cybersecurity.

OSI Model

1. Physical Layer
 2. Data Link Layer
 3. Network Layer
 4. Transport Layer
 5. Session Layer
 6. Presentation Layer
 7. Application Layer
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TCP/IP Model

The TCP/IP model consists of:

- Network Access Layer
 - Internet Layer
 - Transport Layer
 - Application Layer
-

DNS

Domain Name System (DNS) translates domain names into IP addresses.

Example:

google.com → 142.250.x.x

HTTP and HTTPS

HTTP

- HyperText Transfer Protocol
- Data transmitted in plain text

HTTPS

- Secure version of HTTP
 - Uses SSL/TLS encryption
-

NAT

Network Address Translation allows multiple devices to share a single public IP address.

Subnetting

Subnetting divides large networks into smaller manageable networks.

6. CRYPTOGRAPHY BASICS

Cryptography protects information using mathematical algorithms.

Symmetric Encryption

Uses the same key for encryption and decryption.

Examples:

- AES
- DES

Asymmetric Encryption

Uses public and private keys.

Examples:

- RSA
 - ECC
-

Hashing Algorithms

MD5

Produces a 128-bit hash value.

SHA-256

Produces a 256-bit secure hash.

Command Practiced:

```
echo "Hello ApexPlanet" > msg.txt  
sha256sum msg.txt
```

SSL/TLS

SSL/TLS provides secure communication over the internet.

Used in:

```
https://
```

websites.

7. CYBER SECURITY TOOLS

Nmap

Used for:

- Host Discovery
- Port Scanning
- Service Detection

Command:

```
nmap -sV 192.168.56.101
```

Wireshark

Used for:

- Packet Capture
 - Traffic Analysis
 - Protocol Inspection
-

Burp Suite

Used for:

- Web Security Testing
 - HTTP Request Interception
 - Vulnerability Assessment
-

Netcat

Used for:

- Port Testing
- Network Communication
- Banner Grabbing

Command:

```
nc -h
```

8. LAB ENVIRONMENT SETUP

Oracle VirtualBox Installation

Oracle VirtualBox was installed to create virtual machines.

Kali Linux Installation

Kali Linux was installed successfully and configured as the attacker machine.

Metasploitable2 Installation

Metasploitable2 was imported and configured as the target machine.

Network Configuration

Kali Linux

- Adapter 1: NAT
- Adapter 2: Host-Only Adapter

Metasploitable2

- Adapter 1: NAT
 - Adapter 2: Host-Only Adapter
-

IP Verification

Kali Linux

```
192.168.56.102
```

Metasploitable2

```
192.168.56.101
```

9. HOST DISCOVERY AND SERVICE ENUMERATION

Host Discovery

Command:

```
sudo nmap -sn 192.168.56.0/24
```

Result:

```
192.168.56.101
192.168.56.102
```

Service Enumeration

Command:

```
sudo nmap -sV 192.168.56.101
```

Open Services

- FTP
 - SSH
 - Telnet
 - SMTP
 - DNS
 - HTTP
 - SMB
 - MySQL
 - PostgreSQL
 - IRC
-

10. RESULTS

Successfully:

- Installed Kali Linux
 - Installed Metasploitable2
 - Configured Host-Only Networking
 - Verified Connectivity
 - Performed Host Discovery
 - Enumerated Services
 - Explored Wireshark
 - Explored Burp Suite
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11. LEARNING OUTCOMES

Through this task, I learned:

- Cyber Security Fundamentals
- CIA Triad
- Cyber Threats
- Linux Commands
- Networking Concepts

- Cryptography Basics
 - Virtual Machine Setup
 - Host Discovery
 - Service Enumeration
 - Cyber Security Tool Usage
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12. CONCLUSION

Task 1 successfully established the theoretical and practical foundation required for cybersecurity learning. The project covered cybersecurity concepts, Linux fundamentals, networking basics, cryptography, and security tools. A virtual penetration testing lab was created using Kali Linux and Metasploitable2, and reconnaissance activities were performed using Nmap.

This task provided valuable hands-on experience and prepared the environment for future penetration testing and vulnerability assessment tasks.

13. REFERENCES

1. Kali Linux Documentation
2. Oracle VirtualBox Documentation
3. Nmap Documentation
4. Wireshark Documentation
5. Burp Suite Documentation
6. Metasploitable2 Documentation
7. ApexPlanet Cyber Security Internship Guide